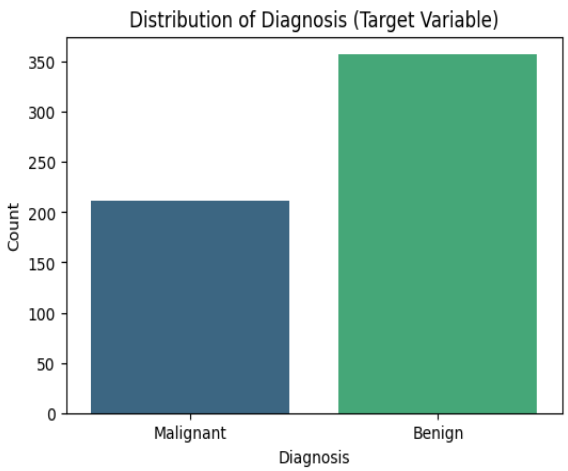
Breast Cancer Prediction Model Evaluation Report

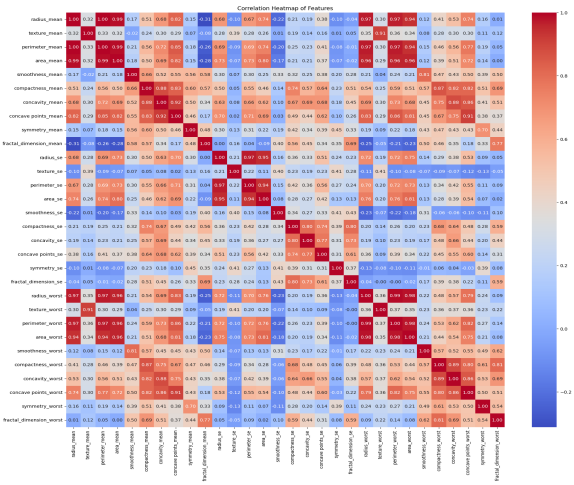
This report summarizes the performance of various machine learning models trained on the Breast Cancer Wisconsin Diagnostic dataset after PCA dimensionality reduction. It also includes key exploratory data analysis plots and model evaluation results.

Exploratory Data Analysis Plots

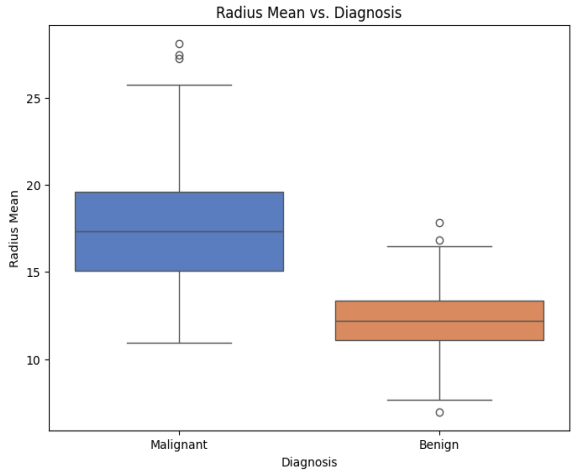
Diagnosis Distribution



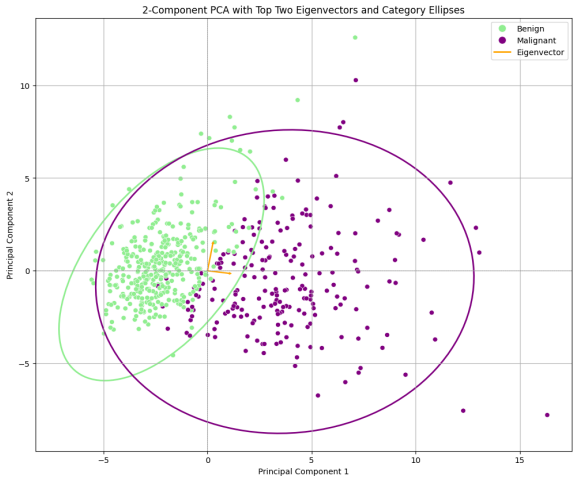
Correlation Heatmap



Radius Mean vs. Diagnosis



PCA with Eigenvectors

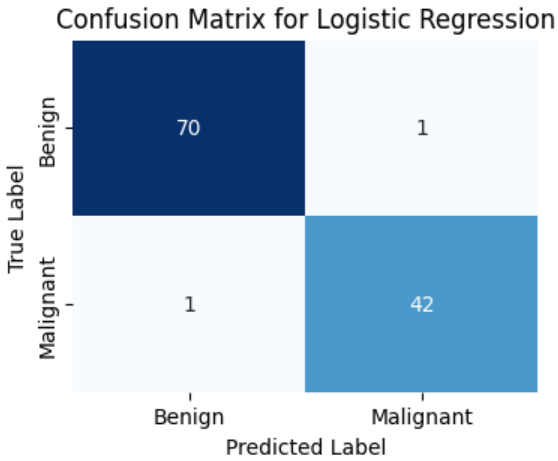


Model Performance Summary

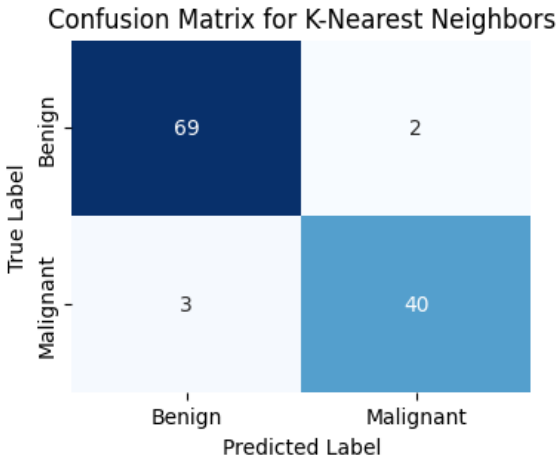
Model	Accuracy	Precision	Recall	F1-Score	Training Time (s)
Logistic Regression	0.9825	0.9767	0.9767	0.9767	0.0078
K-Nearest Neighbors	0.9561	0.9524	0.9302	0.9412	0.0027
Decision Tree	0.9474	0.8936	0.9767	0.9333	0.0086
Random Forest	0.9474	0.9302	0.9302	0.9302	0.2143
Support Vector Machine	0.9737	0.9762	0.9535	0.9647	0.0074

Confusion Matrices

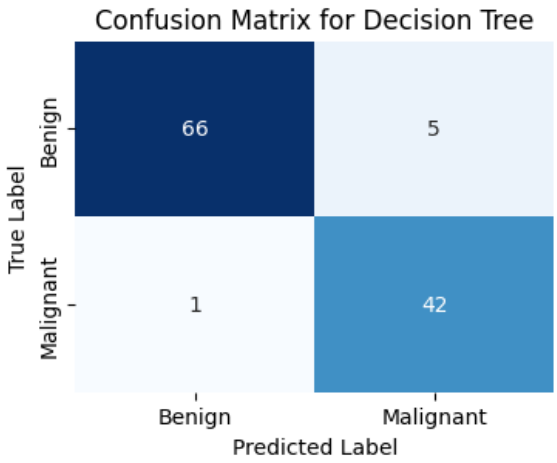
Logistic Regression



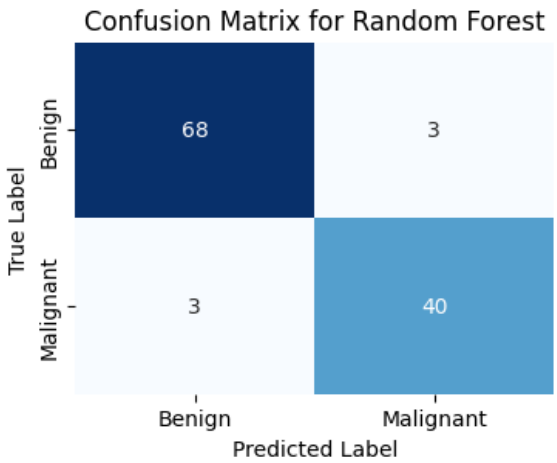
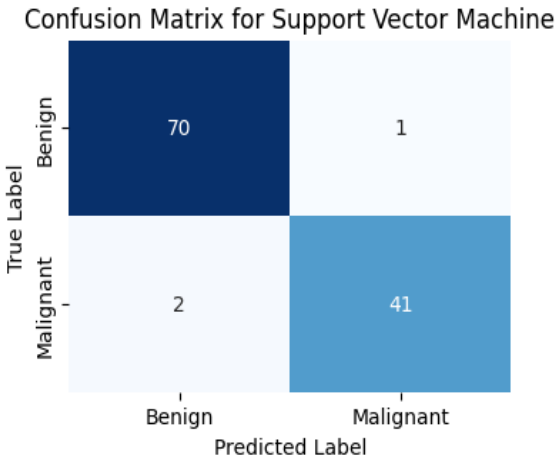
K-Nearest Neighbors



Decision Tree



Random Forest



Support Vector Machine